LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Easy out / Chisel out of rounded screws on aircraft structure, Stop drilling of cracks
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Easy out / Chisel out of rounded screws on aircraft structure, Stop drilling of cracks
5. Specific Objectives	31.1.3.1 Trainee will be able to perform various repair works on aircraft structure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration and supervised practice of Easy Out and Chisel Out techniques on simulated rounded screws; Safety precautions regarding tool handling and structural integrity; Instructor supervision during drilling and extraction processes
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the common causes of rounded screw heads in aircraft structures?
(b) Why is it critical to stop drilling cracks immediately upon detection?
(c) List three tools used for extracting rounded fasteners.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Identification of rounded screw heads • Selection of Easy Out extraction tools • Drilling technique for Easy Out installation • Extraction procedure for Easy Out • Post-extraction hole repair methods	L & Ds	PPT, IB	15 min
(b)	• Assessment of crack dimensions • Stop drilling procedure execution • Drilling sequence for crack arrest • Use of drill bits for crack stopping • Safety measures during crack drilling	L & Ds	PPT, PDF	20 min
(c)	• Review of Easy Out installation steps • Review of crack stop drilling steps • Common errors in fastener extraction • Common errors in crack drilling • Corrective actions for identified errors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step process for installing an Easy Out in a rounded screw hole.
(b) Describe the correct procedure to stop drilling a crack in an aircraft structure.

12. Home Assignment and Follow Up	Prepare a brief report on the tools and materials required for the next lesson on Butt/Lap joints, referencing LP 06 syllabus.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Making of different jobs using Sheet metal i.e Butt/Lap joints	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min